# Distributional Split Criteria for Random Forests: Extensions, Shrinkage, and the Robustness of Mean Splitting

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**Abstract**

Distributional random forests replace mean-based CART splitting with criteria that compare the full conditional response distribution in candidate children. We implement and systematically study a family of such criteria inside a single honest-forest implementation: isotropic random-Fourier-feature maximum mean discrepancy (MMD), an anisotropic diagonal-bandwidth variant, an adaptive per-split frequency-selection variant, and a non-kernel sliced-Wasserstein criterion, together with post-hoc kernel-mean shrinkage of the forest weights. Using paired-seed comparisons across synthetic quantile mechanisms, real univariate benchmarks, a California-housing subsample curve, and multivariate synthetic and real responses, we characterize where each extension pays. Three findings recur. First, among distributional criteria ordinary isotropic MMD is already close to best in class: the anisotropic, adaptive-frequency, and sliced-Wasserstein extensions, and post-hoc shrinkage, do not systematically improve on it. Second, on scalar tabular regression mean-based CART splitting remains the robust default and wins many cells. Third, multivariate responses are the regime where distributional splitting clearly earns its keep, most sharply on a pure-dependence copula where the energy score separates the criteria even though marginal CRPS does not. The evidence supports a simple allocation story: distributional splitting helps only when non-location structure is both present and estimable; otherwise it dilutes split-selection power away from the mean. All criteria, the honest forest, and the paired-comparison harness are implemented in the open-source `drforest` library, whose Rust-backed split search makes broad criterion sweeps inexpensive.

## 1 Question

Random forests for distributional prediction estimate not just a conditional mean, but a conditional law. Quantile regression forests do this with standard CART tree structure and distributional leaf weights [4]. Distributional random forests instead propose splitting rules that compare the full response distributions in candidate children [1]. This is a natural generalization, especially for multivariate responses where a scalar ordering or separate marginal quantile fits are unsatisfactory.

This note studies that design space directly. Inside one honest-forest implementation we add four distributional split criteria—isotropic MMD, a diagonal-bandwidth anisotropic variant, an adaptive per-split frequency-selection variant, and sliced Wasserstein—and a family of post-hoc shrinkage variants, and compare them by paired seed. Two questions organize the study: *among distributional criteria, does any extension beat the plainest MMD?* and *against mean splitting, when is distributional splitting worth its variance?* The answers are, respectively, "not systematically" and "for multivariate responses and a few structured scalar problems," with CART the robust default elsewhere. We test the multivariate regime explicitly, since it is what motivates distributional forests in the first place.

## 2 Split Criteria as Signal Allocation

For a node with observations $(X_i, Y_i)$, a candidate split $A \mid A^c$ is chosen by maximizing an empirical separation score between the child responses. CART uses the between-group mean separation

$$\frac{n_L n_R}{n^2} \|\bar{Y}_L - \bar{Y}_R\|^2,$$

which is the identity-kernel mean-embedding score. Random Fourier feature MMD splitting replaces $Y$ by a complex feature map $\psi(Y)$ and scores

$$\frac{n_L n_R}{n^2} \left\| \frac{1}{n_L} \sum_{i \in L} \psi(Y_i) - \frac{1}{n_R} \sum_{i \in R} \psi(Y_i) \right\|^2 .$$

With enough features and enough data this score can detect changes beyond the mean. The two scores share an exact finite-sample structure: each is a biased squared mean-embedding statistic whose expectation is the population separation plus an $O(n^{-1})$ noise floor set by the embedded-response covariance (Appendix A). The embedding, not the form of the statistic, is all that changes between CART and MMD.

This suggests the working characterization tested below. CART concentrates on the dominant location signal. MMD spreads sensitivity across location and higher-order distributional structure. At finite node sizes the usefulness of distributional splitting is governed by the kernel-mean contrast relative to the covariance of the embedded response, together with random-feature and split-search noise; universal kernel sensitivity does not imply high split-time power for every alternative. MMD should therefore win only when the additional structure is strong enough, or the sample size large enough, for its split-time signal-to-noise ratio to exceed the cost of estimating it (Appendix A).

We compare five criteria: CART; isotropic Gaussian RFF MMD (`mmd_rff`); diagonal-bandwidth MMD (`anisotropic_mmd`); an overcomplete RFF pool with per-split top-$k$ frequency selection (`adaptive_mmd`); and a non-kernel sliced-Wasserstein criterion.[1]

## 3 Evidence

The experiments were run as paired-seed comparisons: the train/test split, forest seeds, tree hyperparameters, honesty setting, and cutpoint grid were held fixed while varying only the split criterion. Metrics are RMSE for the conditional mean and CRPS / energy score for distributional prediction. In every table a negative difference means the row improves over the reference, and the parenthetical is the seed-level win rate.

**Synthetic diagnostics.** The first set of experiments swept the univariate synthetic quantile mechanisms used in the DRF literature (Table 1). When the conditional law carries deliberately strong non-location structure, distributional splitting pays: on `paper_quantile_2` and `paper_quantile_3` with honesty 0.5, every distributional criterion improves CRPS and energy over CART with a 100% win rate while leaving RMSE essentially unchanged. On the more location-driven `paper_quantile_1` the comparison is a wash. Honesty is a visible lever in its own right: the gap between criteria at honesty 0 largely collapses once sample splitting is turned on.

[1] Diagonal-bandwidth MMD is reported only for multivariate responses. On a one-dimensional response it is not a distinct estimator: the coordinatewise median bandwidth coincides with the Euclidean median, and the length-one frequency draw is the same realization as the isotropic scalar-scale draw, so `anisotropic_mmd` reproduces `mmd_rff` bit-for-bit and is omitted from the scalar tables.

Table 1: Paired differences (criterion − `cart`). Negative means the row improves over the reference; values are mean ± paired SE with seed-level win rate.

| dataset | $n_{tr}$ | honesty | criterion | pairs | ΔRMSE | ΔCRPS | Δenergy |
|---|---|---|---|---|---|---|---|
| paper_quantile_1 | 1500 | 0 | adaptive_mmd | 10 | -0.0069 ± 0.0012 (90%) | -0.0032 ± 0.0006 (100%) | -0.0032 ± 0.0006 (100%) |
| paper_quantile_1 | 1500 | 0 | mmd_rff | 10 | -0.0026 ± 0.0011 (80%) | -0.0006 ± 0.0006 (60%) | -0.0006 ± 0.0006 (60%) |
| paper_quantile_1 | 1500 | 0 | sliced_wasserstein | 10 | -0.0011 ± 0.0008 (60%) | -0.0001 ± 0.0005 (40%) | -0.0001 ± 0.0005 (40%) |
| paper_quantile_1 | 1500 | 0.5 | adaptive_mmd | 10 | +0.0003 ± 0.0006 (20%) | +0.0001 ± 0.0003 (30%) | +0.0001 ± 0.0003 (30%) |
| paper_quantile_1 | 1500 | 0.5 | mmd_rff | 10 | +0.0002 ± 0.0003 (40%) | +0.0001 ± 0.0002 (30%) | +0.0001 ± 0.0002 (30%) |
| paper_quantile_1 | 1500 | 0.5 | sliced_wasserstein | 10 | -0.0000 ± 0.0004 (50%) | -0.0001 ± 0.0002 (50%) | -0.0001 ± 0.0002 (50%) |
| paper_quantile_2 | 1500 | 0 | adaptive_mmd | 10 | -0.0082 ± 0.0018 (100%) | -0.0089 ± 0.0011 (100%) | -0.0089 ± 0.0011 (100%) |
| paper_quantile_2 | 1500 | 0 | mmd_rff | 10 | -0.0026 ± 0.0024 (70%) | -0.0060 ± 0.0014 (90%) | -0.0060 ± 0.0014 (90%) |
| paper_quantile_2 | 1500 | 0 | sliced_wasserstein | 10 | +0.0010 ± 0.0016 (30%) | -0.0054 ± 0.0010 (100%) | -0.0054 ± 0.0010 (100%) |
| paper_quantile_2 | 1500 | 0.5 | adaptive_mmd | 10 | +0.0001 ± 0.0005 (70%) | -0.0105 ± 0.0011 (100%) | -0.0105 ± 0.0011 (100%) |
| paper_quantile_2 | 1500 | 0.5 | mmd_rff | 10 | +0.0003 ± 0.0004 (40%) | -0.0105 ± 0.0009 (100%) | -0.0105 ± 0.0009 (100%) |
| paper_quantile_2 | 1500 | 0.5 | sliced_wasserstein | 10 | +0.0007 ± 0.0006 (40%) | -0.0105 ± 0.0011 (100%) | -0.0105 ± 0.0011 (100%) |
| paper_quantile_3 | 1500 | 0 | adaptive_mmd | 10 | -0.0057 ± 0.0017 (90%) | -0.0043 ± 0.0015 (80%) | -0.0043 ± 0.0015 (80%) |
| paper_quantile_3 | 1500 | 0 | mmd_rff | 10 | -0.0018 ± 0.0015 (80%) | -0.0023 ± 0.0012 (80%) | -0.0023 ± 0.0012 (80%) |
| paper_quantile_3 | 1500 | 0 | sliced_wasserstein | 10 | +0.0015 ± 0.0015 (40%) | -0.0021 ± 0.0009 (70%) | -0.0021 ± 0.0009 (70%) |
| paper_quantile_3 | 1500 | 0.5 | adaptive_mmd | 10 | +0.0007 ± 0.0005 (30%) | -0.0034 ± 0.0008 (100%) | -0.0034 ± 0.0008 (100%) |
| paper_quantile_3 | 1500 | 0.5 | mmd_rff | 10 | +0.0003 ± 0.0004 (50%) | -0.0033 ± 0.0006 (100%) | -0.0033 ± 0.0006 (100%) |
| paper_quantile_3 | 1500 | 0.5 | sliced_wasserstein | 10 | +0.0004 ± 0.0003 (30%) | -0.0030 ± 0.0004 (100%) | -0.0030 ± 0.0004 (100%) |

**Real univariate data.** On real univariate benchmarks CART-style splitting is again the stronger default (Table 2). CART wins `wine_quality_white` and `kin8nm` against the MMD family, and `diabetes` ($n = 300$) is too noisy to separate the criteria. The exception worth isolating is California housing, where the MMD family beats CART consistently (isotropic MMD improves CRPS by $\approx 0.003$ at a 100% win rate, and adaptive MMD roughly doubles that). That case confounds two explanations: the dataset is large, and housing prices may contain estimable heteroskedastic structure. A subsample experiment over $n \in \{2000, 4000, 8000, 16000\}$ separates these readings (Figure 1). The paired MMD−CART advantage is already present at small $n$ and is sustained as $n$ grows, which favors the strong-distributional-structure reading over a pure estimability threshold.

Table 2: Paired differences (criterion − `cart`). Negative means the row improves over the reference; values are mean ± paired SE with seed-level win rate.

| dataset | $n_{tr}$ | honesty | criterion | pairs | ΔRMSE | ΔCRPS | Δenergy |
|---|---|---|---|---|---|---|---|
| california_housing | 3000 | 0 | adaptive_mmd | 10 | -0.0051 ± 0.0019 (80%) | -0.0053 ± 0.0008 (100%) | -0.0053 ± 0.0008 (100%) |
| california_housing | 3000 | 0 | mmd_rff | 10 | -0.0032 ± 0.0006 (90%) | -0.0026 ± 0.0003 (100%) | -0.0026 ± 0.0003 (100%) |
| california_housing | 3000 | 0 | sliced_wasserstein | 10 | -0.0005 ± 0.0008 (70%) | -0.0008 ± 0.0004 (80%) | -0.0008 ± 0.0004 (80%) |
| california_housing | 3000 | 0.5 | adaptive_mmd | 10 | -0.0075 ± 0.0014 (90%) | -0.0055 ± 0.0007 (100%) | -0.0055 ± 0.0007 (100%) |
| california_housing | 3000 | 0.5 | mmd_rff | 10 | -0.0045 ± 0.0006 (100%) | -0.0029 ± 0.0003 (100%) | -0.0029 ± 0.0003 (100%) |
| california_housing | 3000 | 0.5 | sliced_wasserstein | 10 | -0.0017 ± 0.0006 (80%) | -0.0010 ± 0.0003 (80%) | -0.0010 ± 0.0003 (80%) |
| diabetes | 300 | 0 | adaptive_mmd | 10 | +0.1814 ± 0.1581 (50%) | +0.2474 ± 0.0867 (20%) | +0.2474 ± 0.0867 (20%) |
| diabetes | 300 | 0 | mmd_rff | 10 | +0.0565 ± 0.0947 (50%) | +0.1077 ± 0.0572 (30%) | +0.1077 ± 0.0572 (30%) |
| diabetes | 300 | 0 | sliced_wasserstein | 10 | +0.0831 ± 0.1027 (50%) | +0.0811 ± 0.0684 (30%) | +0.0811 ± 0.0684 (30%) |
| diabetes | 300 | 0.5 | adaptive_mmd | 10 | +0.1425 ± 0.0994 (30%) | +0.1065 ± 0.0549 (30%) | +0.1065 ± 0.0549 (30%) |
| diabetes | 300 | 0.5 | mmd_rff | 10 | +0.0222 ± 0.0595 (50%) | +0.0176 ± 0.0363 (50%) | +0.0176 ± 0.0363 (50%) |
| diabetes | 300 | 0.5 | sliced_wasserstein | 10 | +0.0401 ± 0.0655 (40%) | +0.0220 ± 0.0368 (50%) | +0.0220 ± 0.0368 (50%) |
| kin8nm | 3000 | 0 | adaptive_mmd | 10 | +0.0069 ± 0.0003 (0%) | +0.0031 ± 0.0001 (0%) | +0.0031 ± 0.0001 (0%) |
| kin8nm | 3000 | 0 | mmd_rff | 10 | +0.0014 ± 0.0002 (10%) | +0.0005 ± 0.0001 (10%) | +0.0005 ± 0.0001 (10%) |
| kin8nm | 3000 | 0 | sliced_wasserstein | 10 | -0.0016 ± 0.0002 (100%) | -0.0008 ± 0.0001 (100%) | -0.0008 ± 0.0001 (100%) |
| kin8nm | 3000 | 0.5 | adaptive_mmd | 10 | +0.0035 ± 0.0003 (0%) | +0.0014 ± 0.0001 (0%) | +0.0014 ± 0.0001 (0%) |
| kin8nm | 3000 | 0.5 | mmd_rff | 10 | +0.0007 ± 0.0001 (0%) | +0.0001 ± 0.0001 (40%) | +0.0001 ± 0.0001 (40%) |
| kin8nm | 3000 | 0.5 | sliced_wasserstein | 10 | -0.0006 ± 0.0001 (100%) | -0.0004 ± 0.0000 (100%) | -0.0004 ± 0.0000 (100%) |
| wine_quality_white | 3000 | 0 | adaptive_mmd | 10 | +0.0191 ± 0.0015 (0%) | +0.0075 ± 0.0009 (0%) | +0.0075 ± 0.0009 (0%) |
| wine_quality_white | 3000 | 0 | mmd_rff | 10 | +0.0068 ± 0.0009 (0%) | +0.0021 ± 0.0005 (10%) | +0.0021 ± 0.0005 (10%) |
| wine_quality_white | 3000 | 0 | sliced_wasserstein | 10 | +0.0072 ± 0.0010 (0%) | +0.0024 ± 0.0005 (10%) | +0.0024 ± 0.0005 (10%) |
| wine_quality_white | 3000 | 0.5 | adaptive_mmd | 10 | +0.0073 ± 0.0009 (0%) | +0.0035 ± 0.0005 (0%) | +0.0035 ± 0.0005 (0%) |
| wine_quality_white | 3000 | 0.5 | mmd_rff | 10 | +0.0026 ± 0.0005 (0%) | +0.0011 ± 0.0002 (10%) | +0.0011 ± 0.0002 (10%) |
| wine_quality_white | 3000 | 0.5 | sliced_wasserstein | 10 | +0.0043 ± 0.0006 (0%) | +0.0023 ± 0.0003 (0%) | +0.0023 ± 0.0003 (0%) |

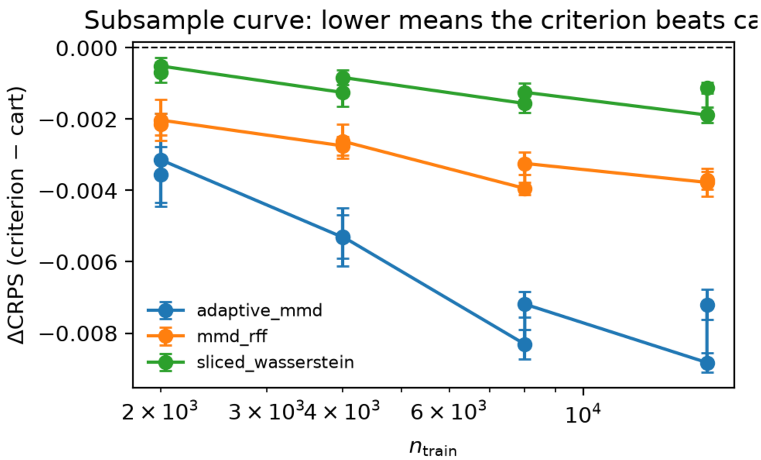


Figure 1: California housing subsample curve: paired CRPS difference (criterion − CART) versus $n_{\text{train}}$. Points below zero mean the criterion beats CART. The MMD family is below CART across the whole range rather than only at large $n$.

**Do the MMD refinements help?** Table 3 references plain isotropic MMD rather than CART and asks whether any refinement is a better distributional baseline. None is. Adaptive frequency selection helps on California but hurts on `kin8nm` and `wine`; sliced Wasserstein helps on `kin8nm` but loses elsewhere; on the multivariate cells the refinements scatter around zero relative to `mmd_rff` with no consistent sign. This is the evidence that ordinary RFF-MMD is already the right distributional default: the extra machinery does not move the frontier. Adaptive-frequency MMD in particular was included as a rescue attempt—if uniform RFF averaging were hiding useful signal, per-split selection should expose it—and it does not change the conclusion, which points away from a simple tuning failure and toward the finite-sample allocation explanation.

**Multivariate responses.** The regime that most naturally motivates distributional forests is multivariate response, and here distributional splitting clearly earns its keep (Table 4). On the building-energy real benchmark (`enb`, $d = 2$) and on two synthetic multivariate mechanisms, the MMD family beats CART on the energy score with high win rates. The cleanest case is `paper_copula`, a pure-dependence shift with little marginal mean signal: every distributional criterion improves the energy score over CART by a wide margin at a 100% win rate, while CRPS and RMSE—which see the marginals only—barely move. This is exactly the structure CART cannot see and a distributional criterion can.

## 4 Characterization

The empirical pattern can be summarized as follows.

> Distributional splitting helps when non-location conditional structure is present and estimable at node scale. For scalar responses in mean-dominated finite samples, CART-style splitting is the better default; among distributional criteria, ordinary isotropic MMD is already close to best in class, and the anisotropic, adaptive, and sliced-Wasserstein refinements do not systematically improve on it. For multivariate responses,

Table 3: Paired differences (criterion − `mmd_rff`). Negative means the row improves over the reference; values are mean ± paired SE with seed-level win rate.

| dataset | $n_{\mathrm{tr}}$ | honesty | criterion | pairs | ΔRMSE | ΔCRPS | Δenergy |
|---|---|---|---|---|---|---|---|
| california_housing | 3000 | 0 | adaptive_mmd | 10 | -0.0020 ± 0.0015 (60%) | -0.0028 ± 0.0007 (100%) | -0.0028 ± 0.0007 (100%) |
| california_housing | 3000 | 0 | cart | 10 | +0.0032 ± 0.0006 (10%) | +0.0026 ± 0.0003 (0%) | +0.0026 ± 0.0003 (0%) |
| california_housing | 3000 | 0 | sliced_wasserstein | 10 | +0.0026 ± 0.0011 (30%) | +0.0017 ± 0.0005 (10%) | +0.0017 ± 0.0005 (10%) |
| california_housing | 3000 | 0.5 | adaptive_mmd | 10 | -0.0031 ± 0.0011 (80%) | -0.0026 ± 0.0006 (90%) | -0.0026 ± 0.0006 (90%) |
| california_housing | 3000 | 0.5 | cart | 10 | +0.0045 ± 0.0006 (0%) | +0.0029 ± 0.0003 (0%) | +0.0029 ± 0.0003 (0%) |
| california_housing | 3000 | 0.5 | sliced_wasserstein | 10 | +0.0027 ± 0.0010 (30%) | +0.0019 ± 0.0005 (20%) | +0.0019 ± 0.0005 (20%) |
| diabetes | 300 | 0 | adaptive_mmd | 10 | +0.1249 ± 0.1012 (40%) | +0.1397 ± 0.0634 (30%) | +0.1397 ± 0.0634 (30%) |
| diabetes | 300 | 0 | cart | 10 | -0.0565 ± 0.0947 (50%) | -0.1077 ± 0.0572 (70%) | -0.1077 ± 0.0572 (70%) |
| diabetes | 300 | 0 | sliced_wasserstein | 10 | +0.0266 ± 0.1016 (70%) | -0.0266 ± 0.0641 (70%) | -0.0266 ± 0.0641 (70%) |
| diabetes | 300 | 0.5 | adaptive_mmd | 10 | +0.1203 ± 0.0578 (30%) | +0.0890 ± 0.0323 (40%) | +0.0890 ± 0.0323 (40%) |
| diabetes | 300 | 0.5 | cart | 10 | -0.0222 ± 0.0595 (50%) | -0.0176 ± 0.0363 (50%) | -0.0176 ± 0.0363 (50%) |
| diabetes | 300 | 0.5 | sliced_wasserstein | 10 | +0.0178 ± 0.0316 (40%) | +0.0045 ± 0.0233 (50%) | +0.0045 ± 0.0233 (50%) |
| enb | 576 | 0 | adaptive_mmd | 5 | -0.0034 ± 0.0018 (80%) | -0.0017 ± 0.0010 (80%) | -0.0023 ± 0.0014 (80%) |
| enb | 576 | 0 | anisotropic_mmd | 5 | -0.0021 ± 0.0025 (60%) | -0.0010 ± 0.0010 (40%) | -0.0013 ± 0.0015 (60%) |
| enb | 576 | 0 | cart | 5 | +0.0027 ± 0.0027 (40%) | +0.0017 ± 0.0011 (20%) | +0.0024 ± 0.0017 (20%) |
| enb | 576 | 0 | sliced_wasserstein | 5 | +0.0090 ± 0.0032 (0%) | +0.0029 ± 0.0013 (20%) | +0.0040 ± 0.0018 (20%) |
| enb | 576 | 0.5 | adaptive_mmd | 5 | -0.0064 ± 0.0044 (60%) | -0.0025 ± 0.0016 (80%) | -0.0036 ± 0.0024 (100%) |
| enb | 576 | 0.5 | anisotropic_mmd | 5 | -0.0007 ± 0.0016 (60%) | -0.0001 ± 0.0004 (40%) | -0.0004 ± 0.0006 (60%) |
| enb | 576 | 0.5 | cart | 5 | +0.0071 ± 0.0037 (20%) | +0.0027 ± 0.0013 (0%) | +0.0036 ± 0.0017 (0%) |
| enb | 576 | 0.5 | sliced_wasserstein | 5 | +0.0089 ± 0.0031 (0%) | +0.0047 ± 0.0010 (0%) | +0.0058 ± 0.0013 (0%) |
| kin8nm | 3000 | 0 | adaptive_mmd | 10 | +0.0056 ± 0.0002 (0%) | +0.0026 ± 0.0001 (0%) | +0.0026 ± 0.0001 (0%) |
| kin8nm | 3000 | 0 | cart | 10 | -0.0014 ± 0.0002 (90%) | -0.0005 ± 0.0001 (90%) | -0.0005 ± 0.0001 (90%) |
| kin8nm | 3000 | 0 | sliced_wasserstein | 10 | -0.0030 ± 0.0003 (100%) | -0.0013 ± 0.0001 (100%) | -0.0013 ± 0.0001 (100%) |
| kin8nm | 3000 | 0.5 | adaptive_mmd | 10 | +0.0028 ± 0.0002 (0%) | +0.0013 ± 0.0001 (0%) | +0.0013 ± 0.0001 (0%) |
| kin8nm | 3000 | 0.5 | cart | 10 | -0.0007 ± 0.0001 (100%) | -0.0001 ± 0.0001 (60%) | -0.0001 ± 0.0001 (60%) |
| kin8nm | 3000 | 0.5 | sliced_wasserstein | 10 | -0.0013 ± 0.0001 (100%) | -0.0005 ± 0.0001 (100%) | -0.0005 ± 0.0001 (100%) |
| paper_copula | 3750 | 0 | adaptive_mmd | 5 | -0.0007 ± 0.0003 (80%) | -0.0003 ± 0.0002 (60%) | -0.0010 ± 0.0005 (100%) |
| paper_copula | 3750 | 0 | anisotropic_mmd | 5 | -0.0008 ± 0.0006 (60%) | +0.0002 ± 0.0004 (40%) | -0.0005 ± 0.0008 (60%) |
| paper_copula | 3750 | 0 | cart | 5 | +0.0009 ± 0.0005 (40%) | +0.0004 ± 0.0003 (40%) | +0.0045 ± 0.0007 (0%) |
| paper_copula | 3750 | 0 | sliced_wasserstein | 5 | +0.0003 ± 0.0007 (40%) | +0.0002 ± 0.0004 (60%) | -0.0002 ± 0.0009 (60%) |
| paper_copula | 3750 | 0.5 | adaptive_mmd | 5 | -0.0001 ± 0.0002 (40%) | -0.0000 ± 0.0001 (40%) | -0.0002 ± 0.0003 (60%) |
| paper_copula | 3750 | 0.5 | anisotropic_mmd | 5 | +0.0003 ± 0.0003 (20%) | +0.0003 ± 0.0001 (20%) | -0.0005 ± 0.0003 (80%) |
| paper_copula | 3750 | 0.5 | cart | 5 | -0.0002 ± 0.0003 (60%) | -0.0001 ± 0.0001 (60%) | +0.0038 ± 0.0003 (0%) |
| paper_copula | 3750 | 0.5 | sliced_wasserstein | 5 | -0.0002 ± 0.0003 (60%) | -0.0000 ± 0.0002 (60%) | -0.0008 ± 0.0004 (80%) |
| paper_heterogeneous_regression | 3750 | 0 | adaptive_mmd | 5 | -0.0008 ± 0.0010 (80%) | +0.0001 ± 0.0008 (60%) | +0.0001 ± 0.0011 (60%) |
| paper_heterogeneous_regression | 3750 | 0 | anisotropic_mmd | 5 | -0.0005 ± 0.0007 (60%) | +0.0005 ± 0.0005 (40%) | +0.0008 ± 0.0007 (20%) |
| paper_heterogeneous_regression | 3750 | 0 | cart | 5 | -0.0012 ± 0.0018 (60%) | +0.0006 ± 0.0008 (20%) | +0.0010 ± 0.0012 (20%) |
| paper_heterogeneous_regression | 3750 | 0 | sliced_wasserstein | 5 | -0.0012 ± 0.0007 (80%) | -0.0005 ± 0.0003 (80%) | -0.0008 ± 0.0005 (80%) |
| paper_heterogeneous_regression | 3750 | 0.5 | adaptive_mmd | 5 | +0.0020 ± 0.0012 (20%) | +0.0004 ± 0.0006 (20%) | +0.0007 ± 0.0010 (20%) |
| paper_heterogeneous_regression | 3750 | 0.5 | anisotropic_mmd | 5 | +0.0007 ± 0.0008 (40%) | +0.0000 ± 0.0003 (40%) | +0.0002 ± 0.0006 (40%) |
| paper_heterogeneous_regression | 3750 | 0.5 | cart | 5 | +0.0023 ± 0.0010 (0%) | +0.0042 ± 0.0004 (0%) | +0.0069 ± 0.0008 (0%) |
| paper_heterogeneous_regression | 3750 | 0.5 | sliced_wasserstein | 5 | +0.0013 ± 0.0005 (20%) | +0.0012 ± 0.0002 (0%) | +0.0019 ± 0.0003 (0%) |
| wine_quality_white | 3000 | 0 | adaptive_mmd | 10 | +0.0123 ± 0.0006 (0%) | +0.0054 ± 0.0004 (0%) | +0.0054 ± 0.0004 (0%) |
| wine_quality_white | 3000 | 0 | cart | 10 | -0.0068 ± 0.0009 (100%) | -0.0021 ± 0.0005 (90%) | -0.0021 ± 0.0005 (90%) |
| wine_quality_white | 3000 | 0 | sliced_wasserstein | 10 | +0.0004 ± 0.0013 (50%) | +0.0003 ± 0.0007 (60%) | +0.0003 ± 0.0007 (60%) |
| wine_quality_white | 3000 | 0.5 | adaptive_mmd | 10 | +0.0047 ± 0.0005 (0%) | +0.0024 ± 0.0003 (0%) | +0.0024 ± 0.0003 (0%) |
| wine_quality_white | 3000 | 0.5 | cart | 10 | -0.0026 ± 0.0005 (100%) | -0.0011 ± 0.0002 (90%) | -0.0011 ± 0.0002 (90%) |
| wine_quality_white | 3000 | 0.5 | sliced_wasserstein | 10 | +0.0017 ± 0.0006 (20%) | +0.0012 ± 0.0004 (20%) | +0.0012 ± 0.0004 (20%) |

where dependence structure is invisible to mean splitting, distributional criteria are the right tool.

This statement is intentionally scoped. It is not a claim that distributional forests are inferior to quantile regression forests, nor that MMD is a poor criterion in general. The core motivation for DRF is multivariate response distributions, where CART/QRF-style univariate orderings are not a complete solution, and the multivariate evidence above supports that motivation directly. The negative result is about the common univariate tabular setting: when the target is scalar and the conditional law is mostly a shifted version of itself, a distributional split criterion is often paying variance to chase weak structure.

Table 4: Paired differences (criterion − `cart`). Negative means the row improves over the reference; values are mean ± paired SE with seed-level win rate.

| dataset | $n_{\mathrm{tr}}$ | honesty | criterion | pairs | Δenergy | ΔCRPS | ΔRMSE |
|---|---|---|---|---|---|---|---|
| enb | 576 | 0 | adaptive_mmd | 5 | -0.0047 ± 0.0026 (100%) | -0.0033 ± 0.0018 (100%) | -0.0062 ± 0.0038 (80%) |
| enb | 576 | 0 | anisotropic_mmd | 5 | -0.0037 ± 0.0024 (80%) | -0.0026 ± 0.0017 (80%) | -0.0048 ± 0.0041 (80%) |
| enb | 576 | 0 | mmd_rff | 5 | -0.0024 ± 0.0017 (80%) | -0.0017 ± 0.0011 (80%) | -0.0027 ± 0.0027 (60%) |
| enb | 576 | 0 | sliced_wasserstein | 5 | +0.0015 ± 0.0022 (40%) | +0.0012 ± 0.0016 (40%) | +0.0063 ± 0.0040 (20%) |
| enb | 576 | 0.5 | adaptive_mmd | 5 | -0.0072 ± 0.0031 (100%) | -0.0052 ± 0.0022 (100%) | -0.0135 ± 0.0067 (100%) |
| enb | 576 | 0.5 | anisotropic_mmd | 5 | -0.0041 ± 0.0019 (80%) | -0.0028 ± 0.0014 (80%) | -0.0078 ± 0.0047 (80%) |
| enb | 576 | 0.5 | mmd_rff | 5 | -0.0036 ± 0.0017 (100%) | -0.0027 ± 0.0013 (100%) | -0.0071 ± 0.0037 (80%) |
| enb | 576 | 0.5 | sliced_wasserstein | 5 | +0.0022 ± 0.0027 (40%) | +0.0020 ± 0.0020 (20%) | +0.0018 ± 0.0058 (60%) |
| paper_copula | 3750 | 0 | adaptive_mmd | 5 | -0.0054 ± 0.0010 (100%) | -0.0007 ± 0.0004 (80%) | -0.0016 ± 0.0008 (100%) |
| paper_copula | 3750 | 0 | anisotropic_mmd | 5 | -0.0050 ± 0.0014 (100%) | -0.0002 ± 0.0006 (60%) | -0.0018 ± 0.0011 (60%) |
| paper_copula | 3750 | 0 | mmd_rff | 5 | -0.0045 ± 0.0007 (100%) | -0.0004 ± 0.0003 (60%) | -0.0009 ± 0.0005 (60%) |
| paper_copula | 3750 | 0 | sliced_wasserstein | 5 | -0.0047 ± 0.0010 (100%) | -0.0002 ± 0.0004 (80%) | -0.0006 ± 0.0006 (80%) |
| paper_copula | 3750 | 0.5 | adaptive_mmd | 5 | -0.0040 ± 0.0004 (100%) | +0.0001 ± 0.0001 (20%) | +0.0001 ± 0.0003 (60%) |
| paper_copula | 3750 | 0.5 | anisotropic_mmd | 5 | -0.0043 ± 0.0002 (100%) | +0.0004 ± 0.0002 (20%) | +0.0004 ± 0.0003 (20%) |
| paper_copula | 3750 | 0.5 | mmd_rff | 5 | -0.0038 ± 0.0003 (100%) | +0.0001 ± 0.0001 (40%) | +0.0002 ± 0.0003 (40%) |
| paper_copula | 3750 | 0.5 | sliced_wasserstein | 5 | -0.0046 ± 0.0002 (100%) | +0.0001 ± 0.0001 (20%) | +0.0000 ± 0.0001 (60%) |
| paper_heterogeneous_regression | 3750 | 0 | adaptive_mmd | 5 | -0.0009 ± 0.0012 (80%) | -0.0006 ± 0.0008 (80%) | +0.0004 ± 0.0022 (40%) |
| paper_heterogeneous_regression | 3750 | 0 | anisotropic_mmd | 5 | -0.0002 ± 0.0010 (40%) | -0.0001 ± 0.0006 (40%) | +0.0008 ± 0.0020 (40%) |
| paper_heterogeneous_regression | 3750 | 0 | mmd_rff | 5 | -0.0010 ± 0.0012 (80%) | -0.0006 ± 0.0008 (80%) | +0.0012 ± 0.0018 (40%) |
| paper_heterogeneous_regression | 3750 | 0 | sliced_wasserstein | 5 | -0.0017 ± 0.0008 (80%) | -0.0011 ± 0.0005 (80%) | -0.0000 ± 0.0015 (40%) |
| paper_heterogeneous_regression | 3750 | 0.5 | adaptive_mmd | 5 | -0.0062 ± 0.0012 (100%) | -0.0039 ± 0.0007 (100%) | -0.0003 ± 0.0018 (20%) |
| paper_heterogeneous_regression | 3750 | 0.5 | anisotropic_mmd | 5 | -0.0067 ± 0.0009 (100%) | -0.0042 ± 0.0005 (100%) | -0.0016 ± 0.0014 (80%) |
| paper_heterogeneous_regression | 3750 | 0.5 | mmd_rff | 5 | -0.0069 ± 0.0008 (100%) | -0.0042 ± 0.0004 (100%) | -0.0023 ± 0.0010 (100%) |
| paper_heterogeneous_regression | 3750 | 0.5 | sliced_wasserstein | 5 | -0.0050 ± 0.0007 (100%) | -0.0031 ± 0.0004 (100%) | -0.0010 ± 0.0007 (80%) |

# 5 Implications

For practice, the recommendation is simple. Use CART-style splitting as the univariate default, then reach for a distributional criterion—ordinary isotropic MMD, without further refinement—when the response is multivariate or the scalar data show clear heteroskedasticity, tail behavior, or non-location structure. When distributional splitting is used, paired-seed comparisons and honesty settings should be reported: in these experiments, honesty can be as important as the criterion itself. All of the above is implemented in the open-source `drforest` library, which provides the honest forest, every criterion compared here, and the post-hoc shrinkage variants behind one interface; its Rust-backed split search ran the full paired sweep in this note at a cost that makes such characterizations routine rather than a project in themselves.

For research, the next useful theory is not a generic dominance theorem. The useful object is a finite-sample comparison of split-score signal-to-noise: mean separation versus kernel mean separation under local alternatives. Appendix A carries this out for a single fixed split and shows that the location-shift mechanism is visible to both criteria, whereas the mean-preserving scale, shape, and dependence mechanisms have zero population mean contrast but positive Gaussian-kernel contrast. This matches the qualitative visibility pattern in the experiments. What remains genuinely hard is lifting this fixed-split calculation to the adaptive maximization over correlated cutpoints, the per-node random-feature resampling, and the forest aggregation that ultimately determines CRPS or energy risk; that is where a sharp threshold statement, as opposed to a mechanism, would have to come from.

# 6 Limitations

The scalar experiments compare CART-style splitting to MMD splitting inside the same forest implementation; a full external QRF package comparison is a separate baseline. The multivariate evidence, while decisive on the energy score, uses a small number of mechanisms and one real benchmark, and a broader multivariate suite would strengthen the positive side of the characterization.

# A Finite-Node Analysis of Split Scores

This appendix collects the tractable theory behind the signal-allocation characterization. Throughout we fix a single node and a single candidate split $L \mid R$, with $n$ observations, child sizes $n_L, n_R$, and balance $q = n_L/n$. Conditional on the child memberships we treat the within-child responses as i.i.d. draws from the two child laws; this is a fixed-split statement, not a statement about the cutpoint actually selected by maximizing the score, and the gap between the two is part of what makes a sharp threshold result hard (Remark 1).

For the RFF criterion, all statements below condition on the sampled frequencies and on the forest-level bandwidth. We identify the complex feature vector in $\mathbb{C}^B$ with its real representation in $\mathbb{R}^{2B}$ when invoking covariances and Gaussian limits.

Let $g(\cdot)$ denote the response embedding scored by the criterion: $g(Y) = Y$ for CART, and $g(Y) = B^{-1/2}\psi_B(Y)$ for the normalized $B$-feature RFF map, so that $\|g\|$ is bounded and the MMD score below is the implemented one. Write $\mu_{g,L}, \mu_{g,R}$ for the child means of $g(Y)$, $C_L, C_R$ for its child covariance operators, and

$$\delta_g = \mu_{g,L} - \mu_{g,R}, \qquad G_g = q(1-q)\,\|\delta_g\|^2.$$

Both criteria score the same biased squared mean-embedding statistic, differing only through $g$:

$$\widehat{G}_g = q(1-q)\,\big\|\bar{g}_L - \bar{g}_R\big\|^2, \qquad \bar{g}_L = \tfrac{1}{n_L}\sum_{i\in L} g(Y_i), \quad \bar{g}_R = \tfrac{1}{n_R}\sum_{i\in R} g(Y_i).$$

## A.1 Exact finite-sample expectation

**Proposition 1.** *Conditional on the child sizes,*

$$\mathbb{E}\big[\widehat{G}_g\big] = G_g + \frac{(1-q)\operatorname{tr}(C_L) + q\operatorname{tr}(C_R)}{n}.$$

*Proof.* The increment $\bar{g}_L - \bar{g}_R$ has mean $\delta_g$ and covariance $C_L/n_L + C_R/n_R$, so $\mathbb{E}\|\bar{g}_L - \bar{g}_R\|^2 = \|\delta_g\|^2 + \operatorname{tr}(C_L)/n_L + \operatorname{tr}(C_R)/n_R$. Multiplying by $q(1-q)$ and using $q(1-q)/n_L = (1-q)/n$ and $q(1-q)/n_R = q/n$ gives the claim. □

Three consequences are worth stating. The population signal is $\|\delta_g\|^2$; the score carries an $O(n^{-1})$ noise floor; and when the two children share a law ($C_L = C_R = C$) that floor is $\operatorname{tr}(C)/n$ regardless of split balance. CART and MMD obey the same decomposition—only the embedding $g$, and hence $\operatorname{tr}(C_\bullet)$, changes. This decomposition does not by itself explain the honesty comparison: honesty reduces the structure-sample size used to score splits while separating structure selection from leaf estimation.

## A.2 Local quadratic-form limit and signal-to-noise

Assume finite second moments, $q_n \to q \in (0,1)$, and convergence of the child covariance operators. Let $W_g = C_L/q + C_R/(1-q)$, the limiting $n$-scaled covariance of the increment. A multivariate central limit theorem gives

$$\sqrt{n}\,\big((\bar{g}_L - \bar{g}_R) - \delta_g\big) \;\Rightarrow\; \mathcal{N}(0, W_g).$$

Under the local alternative $\delta_{g,n} = h_g/\sqrt{n}$,

$$n\,\widehat{G}_g \;\Rightarrow\; q(1-q)\,\|Z_g + h_g\|^2, \qquad Z_g \sim \mathcal{N}(0, W_g).$$

Setting $h_g = 0$ recovers the null: for CART ($g(Y) = Y$, scalar) this is a scaled $\chi^2_1$, whereas for the finite-$B$ RFF embedding it is a weighted central quadratic form in Gaussians. Under $h_g \neq 0$ the corresponding local limit is generally noncentral. Under the Gaussian approximation a natural fixed-split signal-to-noise ratio follows from $\mathbb{E}\|Z\|^2 = \|\delta\|^2 + \operatorname{tr}\Sigma$ and $\operatorname{Var}\|Z\|^2 = 2\operatorname{tr}(\Sigma^2) + 4\delta^\top\Sigma\delta$ with $\Sigma = W_g/n$:

$$\mathrm{SNR}_g \;\approx\; \frac{\|\delta_g\|^2}{\sqrt{4\langle\delta_g, W_g\,\delta_g\rangle/n + 2\|W_g\|^2_{\mathrm{HS}}/n^2}}.$$

This formalizes signal allocation correctly: power depends on the embedding contrast $\|\delta_g\|^2$ relative to the covariance geometry $W_g$ of the embedded response, not on a bare count of feature coordinates. A normalized bounded kernel does not incur an automatic dimension penalty.

### A.3 Two-regime visibility and the synthetic mechanisms

**Proposition 2.** *Suppose that, within the node, each side is a mixture of the same two fixed laws $P_0, P_1$ with regime proportions $\alpha_L, \alpha_R$. Writing $m_{g,i} = \mathbb{E}_{P_i}[g]$,*

$$\delta_g = (\alpha_L - \alpha_R)\,(m_{g,1} - m_{g,0}), \qquad G_g = q(1-q)\,(\alpha_L - \alpha_R)^2\,\|m_{g,1} - m_{g,0}\|^2.$$

The cutpoint-dependent factor $q(1-q)(\alpha_L - \alpha_R)^2$ is shared by every embedding; criteria differ only through the constant $\|m_{g,1} - m_{g,0}\|^2$. Hence every embedding that can see $P_0 \neq P_1$ induces the *same* population ranking over cutpoints, up to that constant. CART's constant is $\|\mathbb{E}_{P_1}Y - \mathbb{E}_{P_0}Y\|^2$ and vanishes exactly when the two regimes share a mean. On `paper_quantile_1`, a pure location shift, CART and MMD share population signal and any difference is finite-sample. On `paper_quantile_2` (equal means, scale $1 \to 2$), `paper_quantile_3` (Gaussian vs. exponential, both unit mean), CART's population contrast is exactly zero while the Gaussian-kernel contrast is positive.

The copula mechanism is not a two-regime model: its conditional law varies continuously with $X_1 \sim \mathrm{Unif}(0,1)$. Nevertheless, any interior cut $X_1 = t$ has zero child-mean contrast, while for distinct response coordinates $j \neq k$ the child covariances are $\mathrm{Cov}(Y_j, Y_k \mid X_1 \le t) = t/2$ and $\mathrm{Cov}(Y_j, Y_k \mid X_1 > t) = (1+t)/2$. Thus the joint child laws differ, and their exact Gaussian-kernel MMD is positive.

Table 5: Population split visibility for the synthetic diagnostics. The quantile mechanisms change regime at $X_1 = 0$; the copula result holds at any interior cut $X_1 = t \in (0,1)$. "Mean" is the CART contrast $\|\delta_Y\|$ and "kernel" is the exact Gaussian-MMD contrast.

| mechanism | distributional change | mean signal | kernel signal |
|---|---|---|---|
| `paper_quantile_1` | $\mathcal{N}(0.8,1)$ vs. $\mathcal{N}(0,1)$ | yes | yes |
| `paper_quantile_2` | $\mathcal{N}(0,4)$ vs. $\mathcal{N}(0,1)$ | 0 | yes |
| `paper_quantile_3` | $\mathrm{Exp}(1)$ vs. $\mathcal{N}(1,1)$ | 0 | yes |
| `paper_copula` | corr. $X_1$, fixed $\mathcal{N}(0,1)$ marginals | 0 | joint only |

For the Gaussian kernel $k(y, y') = \exp\{-(y-y')^2/2\sigma^2\}$ the first two scalar contrasts are closed-form, using $\mathbb{E}_{\mathcal{N}(m,v)} \exp\{-U^2/2\sigma^2\} = \frac{\sigma}{\sqrt{\sigma^2+v}} \exp\{-m^2/2(\sigma^2+v)\}$:

$$\mathrm{MMD}^2\{\mathcal{N}(0,1), \mathcal{N}(0.8,1)\} = \frac{2\sigma}{\sqrt{\sigma^2+2}}\left[1 - \exp\left\{-\frac{0.8^2}{2(\sigma^2+2)}\right\}\right],$$

$$\mathrm{MMD}^2\{\mathcal{N}(0,1), \mathcal{N}(0,4)\} = \frac{\sigma}{\sqrt{\sigma^2+2}} + \frac{\sigma}{\sqrt{\sigma^2+8}} - \frac{2\sigma}{\sqrt{\sigma^2+5}},$$

which are exactly the `paper_quantile_1` and `paper_quantile_2` splits. Both are strictly positive for every bandwidth $\sigma$: the first confirms that both criteria see the location shift, while the second retains kernel signal despite a zero CART contrast.

### A.4 What a uniform bound can and cannot give

Fix the bandwidth and condition on the predictor-derived candidate grid. For a finite candidate set $\mathcal{S}$, let $G_k(s)$ be the exact-kernel population score and $\widehat{G}_B(s)$ the empirical RFF score. Bounded-feature concentration and a union bound over $\mathcal{S}$ yield a uniform control of schematic form

$$\sup_{s\in\mathcal{S}} \left|\widehat{G}_B(s) - G_k(s)\right| \lesssim \sqrt{\frac{\log(|\mathcal{S}|/\alpha)}{n_{\min}}} + \frac{\log(|\mathcal{S}|/\alpha)}{n_{\min}} + \sqrt{\frac{\log(|\mathcal{S}|/\alpha)}{B}},$$

whose three terms are within-node response sampling, the quadratic remainder, and the random-feature approximation [3, 5]. On an event where the right-hand side is at most $\varepsilon$, the empirical maximizer has population regret at most $2\varepsilon$; if the unique population maximizer is separated from every competitor by more than $2\varepsilon$, it is recovered exactly. The same argument gives informative-feature recovery when that margin separates the best informative split from all uninformative splits. The bound captures minimum child size $n_{\min}$, the number of searched cutpoints $|\mathcal{S}|$, and the feature budget $B$. We state it schematically because a sharp constant is not the point and is genuinely harder than the bound suggests (Remark 1).

*Remark* 1. A universal threshold at which MMD splitting overtakes CART is not available without heavily restricting the model. There is no universal ordering: under a fixed-split homoskedastic Gaussian location alternative, the standardized mean difference is the likelihood-ratio statistic and CART is locally optimal. Under mean-preserving variance, shape, or dependence alternatives, CART has zero population contrast while an exact characteristic-kernel MMD has positive contrast; a fixed finite RFF map additionally requires that its sampled frequencies see the alternative. Beyond this, tree building maximizes the score over strongly correlated cutpoints (so fixed-split SNR is not selection probability), the RFF frequencies are resampled per node, and the median bandwidth is estimated once from the training responses and held fixed for the forest. Recursive splitting changes later node laws, and converting root-node selection power into final CRPS or energy risk requires analyzing the adaptive forest partition and its aggregation. The original DRF analysis already supplies the RKHS–CART interpretation and whole-forest consistency [1], and fixed-$B$ RFF cannot support unrestricted consistency claims [2]; the contribution here is the exact finite-node comparison, not a dominance theorem.

## B Shrinkage Frontier

Post-hoc shrinkage of forest weights is a separate lever from the split criterion. Table 6 pairs each shrinkage variant against raw weights within a fixed criterion and forest, so the difference isolates the shrinkage effect. Shrinkage does not change the criterion ranking: marginal shrinkage is at best a small CRPS improvement on one synthetic mechanism, parent shrinkage runs to large intensity and consistently hurts, and the pattern is the same across CART, MMD, and sliced-Wasserstein forests. Shrinkage is not the missing ingredient that would let the MMD variants dominate CART on scalar data.

Table 6: Shrinkage frontier: paired differences (variant − `raw`) within a fixed criterion and forest. Negative means shrinkage improves over raw weights; $\bar{\alpha}$ is the mean shrinkage intensity.

| dataset | criterion | honesty | variant | pairs | $\bar{\alpha}$ | ΔRMSE | ΔCRPS | Δenergy |
|---|---|---|---|---|---|---|---|---|
| enb | cart | 0.5 | marginal_kmse | 5 | 0.003 | -0.0001 ± 0.0008 (40%) | +0.0009 ± 0.0001 (0%) | +0.0012 ± 0.0002 (0%) |
| enb | cart | 0.5 | marginal_stein | 5 | 0.003 | -0.0001 ± 0.0008 (40%) | +0.0009 ± 0.0001 (0%) | +0.0012 ± 0.0002 (0%) |
| enb | cart | 0.5 | parent_kmse | 5 | 0.282 | +0.0472 ± 0.0053 (0%) | +0.0351 ± 0.0017 (0%) | +0.0509 ± 0.0022 (0%) |
| enb | cart | 0.5 | parent_stein | 5 | 0.402 | +0.0612 ± 0.0084 (0%) | +0.0494 ± 0.0029 (0%) | +0.0719 ± 0.0039 (0%) |
| enb | mmd_rff | 0.5 | marginal_kmse | 5 | 0.003 | -0.0001 ± 0.0008 (40%) | +0.0009 ± 0.0001 (0%) | +0.0013 ± 0.0002 (0%) |
| enb | mmd_rff | 0.5 | marginal_stein | 5 | 0.003 | -0.0001 ± 0.0008 (40%) | +0.0009 ± 0.0001 (0%) | +0.0013 ± 0.0002 (0%) |
| enb | mmd_rff | 0.5 | parent_kmse | 5 | 0.286 | +0.0487 ± 0.0044 (0%) | +0.0361 ± 0.0013 (0%) | +0.0522 ± 0.0016 (0%) |
| enb | mmd_rff | 0.5 | parent_stein | 5 | 0.408 | +0.0623 ± 0.0070 (0%) | +0.0507 ± 0.0023 (0%) | +0.0735 ± 0.0030 (0%) |
| enb | sliced_wasserstein | 0.5 | marginal_kmse | 5 | 0.003 | -0.0001 ± 0.0008 (40%) | +0.0009 ± 0.0001 (0%) | +0.0012 ± 0.0002 (0%) |
| enb | sliced_wasserstein | 0.5 | marginal_stein | 5 | 0.003 | -0.0002 ± 0.0008 (40%) | +0.0009 ± 0.0001 (0%) | +0.0012 ± 0.0002 (0%) |
| enb | sliced_wasserstein | 0.5 | parent_kmse | 5 | 0.275 | +0.0495 ± 0.0052 (0%) | +0.0360 ± 0.0015 (0%) | +0.0520 ± 0.0019 (0%) |
| enb | sliced_wasserstein | 0.5 | parent_stein | 5 | 0.396 | +0.0644 ± 0.0081 (0%) | +0.0510 ± 0.0027 (0%) | +0.0740 ± 0.0035 (0%) |
| paper_quantile_2 | cart | 0.5 | marginal_kmse | 5 | 0.316 | -0.0007 ± 0.0002 (100%) | +0.0018 ± 0.0003 (0%) | +0.0018 ± 0.0003 (0%) |
| paper_quantile_2 | cart | 0.5 | marginal_stein | 5 | 0.460 | -0.0010 ± 0.0003 (100%) | +0.0026 ± 0.0005 (0%) | +0.0026 ± 0.0005 (0%) |
| paper_quantile_2 | cart | 0.5 | parent_kmse | 5 | 0.839 | -0.0004 ± 0.0003 (80%) | +0.0009 ± 0.0002 (0%) | +0.0009 ± 0.0002 (0%) |
| paper_quantile_2 | cart | 0.5 | parent_stein | 5 | 1.000 | -0.0004 ± 0.0004 (80%) | +0.0013 ± 0.0003 (0%) | +0.0013 ± 0.0003 (0%) |
| paper_quantile_2 | mmd_rff | 0.5 | marginal_kmse | 5 | 0.059 | -0.0001 ± 0.0001 (80%) | +0.0002 ± 0.0001 (20%) | +0.0002 ± 0.0001 (20%) |
| paper_quantile_2 | mmd_rff | 0.5 | marginal_stein | 5 | 0.066 | -0.0002 ± 0.0001 (80%) | +0.0002 ± 0.0002 (40%) | +0.0002 ± 0.0002 (40%) |
| paper_quantile_2 | mmd_rff | 0.5 | parent_kmse | 5 | 0.903 | -0.0002 ± 0.0005 (60%) | -0.0001 ± 0.0003 (60%) | -0.0001 ± 0.0003 (60%) |
| paper_quantile_2 | mmd_rff | 0.5 | parent_stein | 5 | 1.000 | -0.0003 ± 0.0005 (60%) | -0.0002 ± 0.0003 (60%) | -0.0002 ± 0.0003 (60%) |
| paper_quantile_2 | sliced_wasserstein | 0.5 | marginal_kmse | 5 | 0.057 | -0.0002 ± 0.0001 (80%) | +0.0002 ± 0.0001 (20%) | +0.0002 ± 0.0001 (20%) |
| paper_quantile_2 | sliced_wasserstein | 0.5 | marginal_stein | 5 | 0.064 | -0.0001 ± 0.0000 (80%) | +0.0002 ± 0.0002 (20%) | +0.0002 ± 0.0002 (20%) |
| paper_quantile_2 | sliced_wasserstein | 0.5 | parent_kmse | 5 | 0.904 | -0.0000 ± 0.0003 (60%) | +0.0001 ± 0.0002 (40%) | +0.0001 ± 0.0002 (40%) |
| paper_quantile_2 | sliced_wasserstein | 0.5 | parent_stein | 5 | 1.000 | -0.0000 ± 0.0004 (60%) | +0.0001 ± 0.0003 (40%) | +0.0001 ± 0.0003 (40%) |
| shrinkage_toy | cart | 0.5 | marginal_kmse | 5 | 0.040 | +0.0076 ± 0.0018 (0%) | +0.0038 ± 0.0007 (0%) | +0.0061 ± 0.0011 (0%) |
| shrinkage_toy | cart | 0.5 | marginal_stein | 5 | 0.042 | +0.0081 ± 0.0020 (0%) | +0.0040 ± 0.0007 (0%) | +0.0064 ± 0.0012 (0%) |
| shrinkage_toy | cart | 0.5 | parent_kmse | 5 | 0.379 | +0.0137 ± 0.0036 (0%) | +0.0072 ± 0.0014 (0%) | +0.0135 ± 0.0023 (0%) |
| shrinkage_toy | cart | 0.5 | parent_stein | 5 | 0.586 | +0.0229 ± 0.0053 (0%) | +0.0119 ± 0.0023 (0%) | +0.0223 ± 0.0036 (0%) |
| shrinkage_toy | mmd_rff | 0.5 | marginal_kmse | 5 | 0.039 | +0.0075 ± 0.0019 (0%) | +0.0037 ± 0.0007 (0%) | +0.0060 ± 0.0011 (0%) |
| shrinkage_toy | mmd_rff | 0.5 | marginal_stein | 5 | 0.041 | +0.0079 ± 0.0020 (0%) | +0.0039 ± 0.0007 (0%) | +0.0063 ± 0.0012 (0%) |
| shrinkage_toy | mmd_rff | 0.5 | parent_kmse | 5 | 0.376 | +0.0131 ± 0.0036 (0%) | +0.0068 ± 0.0015 (0%) | +0.0129 ± 0.0024 (0%) |
| shrinkage_toy | mmd_rff | 0.5 | parent_stein | 5 | 0.578 | +0.0216 ± 0.0054 (0%) | +0.0112 ± 0.0024 (0%) | +0.0211 ± 0.0037 (0%) |
| shrinkage_toy | sliced_wasserstein | 0.5 | marginal_kmse | 5 | 0.039 | +0.0076 ± 0.0019 (0%) | +0.0038 ± 0.0007 (0%) | +0.0061 ± 0.0011 (0%) |
| shrinkage_toy | sliced_wasserstein | 0.5 | marginal_stein | 5 | 0.042 | +0.0081 ± 0.0020 (0%) | +0.0040 ± 0.0007 (0%) | +0.0065 ± 0.0012 (0%) |
| shrinkage_toy | sliced_wasserstein | 0.5 | parent_kmse | 5 | 0.383 | +0.0131 ± 0.0034 (0%) | +0.0068 ± 0.0013 (0%) | +0.0130 ± 0.0021 (0%) |
| shrinkage_toy | sliced_wasserstein | 0.5 | parent_stein | 5 | 0.591 | +0.0220 ± 0.0051 (0%) | +0.0114 ± 0.0022 (0%) | +0.0216 ± 0.0034 (0%) |

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